

Microfinance regulation: investigating the interplay between interest rate caps and competition in a financial inclusion context

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Abstract

The microfinance industry has rapidly developed and commercialized since its early days, exacerbating competition and emphasizing the attention paid to profits. As a response, many governments have been tempted to cap lending interest rates, a measure generally considered as socially counterproductive. With a consolidated dataset including 986 microfinance institutions from 73 countries over 2015-2018, we assess the effect of interest rate caps on the average loan size provided by microfinance institutions with fixed-effect regressions. In addition, we use a moderation analysis with several indicators, including the Herfindahl-Hirschman and Lerner indexes, to understand how competition affects this initial relationship. Our findings confirm that institutions facing interest rate caps provide larger loans on average and suggest that competition amplifies this phenomenon.

KEYWORDS: Microfinance, Regulation, Interest rate cap, Competition, Moderation, Outreach

1 Introduction

Microfinance institutions (MFIs) aim at providing financial services to the poor, unbanked people, while simultaneously seeking financial sustainability (Morduch, 1999a). Although marginal in the 70s-80s, the industry has substantially grown and evolved. Today, the number of clients served has significantly risen, the loan portfolio has reached unprecedented levels, the range of services has diversified, and microfinance is now a major sector in many emerging countries. Yet, in some regions, severe crises have tarnished these successes, revealing abusive practices, aggressive competition, mission drift, ethical debates, and over-indebtedness (Armendáriz and Szafarz, 2011; Sandberg, 2012; Schicks and Rosenberg, 2011; Labie et al., 2015).

Looking back on the development of the industry, two essential trends have had a significant influence. On the one hand, while microfinance emerged mostly through Non-Governmental Organizations (NGOs), it has drawn the attention of profit-seeking actors, which nowadays seem to dominate (Gonzalez and Rosenberg, 2006). In addition, microfinance increasingly represents an opportunity for commercial investors, who often pay a large attention to financial performance (Mersland and Urgeghe, 2013). This general commercialization of the industry has led to more and more competitive pressure, with benefits but also worrying effects. While competition may indeed encourage efficiency, reduced lending rates, and innovation, some scholars alerted about undesired effects related to information asymmetry or risk-taking (McIntosh and Wydick, 2005). At the end of the day, the literature highlights ambiguous effects of competition in the financial inclusion context (Assefa et al., 2013).

On the other hand, formal regulation has also intensified (Kirkpatrick and Maimbo, 2002). While microfinance benefited from low constraints in its early days, a consensus has formed around the idea that regulation is essential to make sure MFIs do not leave aside their social goals (Arun, 2005). Yet, regulation does not necessarily entail better financial nor social outcomes as such (Hartarska and Nadolnyak, 2007) and, when maladapted, may even destabilize the industry (Ménard, 1998; Guérin et al., 2018). Particularly, concerned about the potential drawbacks of the quest for profit, more and more governments have been tempted to impose legal limits to the rates charged by MFIs (Christen et al., 2012). Despite their a priori honourable motivations, interest rate caps are however mainly considered as detrimental. Especially, caps could force MFIs to drop their costliest clients, often the poorest and most difficult to reach. Rather than protecting the poor, caps would thus exacerbate financial exclusion. Although caps are debated in the literature, grey areas remain on their effects in microfinance.

In such a context, the goal of this paper is two-fold. First, it aims at taking a step back to study how interest rate caps affect financial inclusion. Using a consolidated dataset of 986 MFIs from 73 countries over 2015-2018, we assess how caps affect the depth of outreach of MFIs. We use the average loan size as a proxy: the smaller the loan, the poorer the clients. Our results show that MFIs facing interest rate caps provide larger loans on average, a sign of the financial exclusion of the poorest. Second, we go one step further by studying how competition alters this initial relationship through a moderation analysis. Using several

indicators, including the Herfindahl-Hirschman and the Lerner indexes, we show that competition exacerbates the detrimental effect of a cap.

Doing so, we bring two main contributions. First, although caps are discussed in the literature, empirical – and even more quantitative – analyses of their effects in microfinance are relatively scarce. Additionally, studies on caps are either conceptual, centred on a specific case study, or constituted of inventories or taxonomies of caps around the world. We provide a more global and quantitative analysis than previous research, thanks to recent data that we collected on the use of caps in microfinance. Second, as competition heavily influences MFIs, regulation may have different effects in different market conditions (Karimu et al., 2019). Yet, studies on regulation in microfinance – and a fortiori on caps – tend to focus on the regulation itself, leaving aside the market circumstances that may condition its outcomes. We contribute to the literature by adopting a systemic approach and studying how competition interacts with regulation to shape MFIs’ behavior.

The paper is structured as follows: Section 2 presents a literature review around regulation, interest rate caps and competition in microfinance. Section 3 describes the data. Section 4 presents the method. Eventually, Section 5 details the results and Section 6 concludes.

2 Regulation and competition in a financial inclusion context

Governmental interventions and competition have traditionally been opposed in the literature. Looking back at Smith’s invisible hand, Friedman’s work on the monetary policy and the Keynesian approach of State intervention, the tradition has been to see both forces as antagonists and to look at them in relatively separate perspectives. This has resulted in a somewhat dichotomic analysis of regulation in the academic research. Focusing on the microfinance industry, the literature suggests that both, regulation and competition, may influence the financial as well as social outcomes of microfinance in various ways.

Regulation, interest rate caps and microfinance

Governments have embodied different roles when intervening in microfinance markets. When the sector emerged, back in the 70s-80s, governments were active in the provision of credit with relaxed terms and subsidized rates but relatively absent regarding regulation and supervision, which MFIs seem to have benefited from. As the sector developed and diversified beyond credit, “microfinance began to attract regulatory concern” (Kirkpatrick and Maimbo, 2002:296). Regulation is now considered as key and influencing the operations of the MFIs (Olivares-Polanco, 2005).

Regulators have adopted diverse approaches, from copy-pasting banking laws to regulating “by exception” or designing specific microfinance regulations. Today, most microfinance industries present at least some form of regulation, and the common view seems to be that there is no unique effective framework fitting all markets but, rather, a bunch of consensual principles (Rosengard, 2011). These principles have particularly been supported by the Consultative Group to Assist the Poor (CGAP), an independent policy and research consortium, and aim at guiding regulators and preventing damaging interventions.

Among these core principles, the CGAP points that interest rate caps do not help the poor as they “can restrict access by making it impossible to serve small or remote borrowers” (Christen et al., 2012:56). In fact, MFIs generally target financially excluded and mostly poor clients, which requires tiny loans. Contrarily to mainstream credit, tiny loans are more expensive to provide, due to a higher cost per unit, even more when considering people from remote rural areas. Furthermore, for serving such clients, microfinance requires a different, costlier business model than what banks generally do, due to economic as well as sociological reasons (Labie, 2004). Therefore, the only way to serve these clients formally and sustainably is to charge rates that may seem excessive if compared with those charged for traditional credits. Yet, concerned about potentially excessive interests, regulators often consider banking rates as a reference point for setting interest rate caps, making it usually impossible for numerous MFIs to provide microcredits to their poorest clients at viable rates (Helms and Reille, 2004; Rosengard, 2011). Moreover, although a cap could in theory be implemented at a sustainable level, the diversity and specificities of MFIs make it particularly arduous to identify such level in practice, and inappropriate to apply an often used one-size-fits-all approach when setting caps (Zetzsche and Dewi, 2018). This way, imposing caps would force MFIs to exclude the clients requiring a higher rate than the limit to cover the cost and/or risk they entail. The costliest clients being the poorest and the ones with the smallest loans, capping MFIs’ rates would end up exacerbating financial exclusion (Hudon, 2007). Empirically, Latortue (2004) and Attuel-Mendès and Ashta (2008) observed respectively in Nicaragua and in France that microborrowers had less access to credit after the imposition of a cap. Campion et al. (2010), studying Latin America and the Caribbean, also suggest that caps damage MFIs’ outreach to the poorest, especially women (being generally poorer) and rural clients (located usually in poorer, remote areas). More recently, Ferrari et al. (2018) observed a substantial increase in average loan sizes in Cambodia after a cap was set under the average market rate.

Although the rationale behind the implications of caps for the financial inclusion of the poor has been debated in the literature (Gonzalez-Vega, 1977; Acclassato, 2008; Miller, 2013; Maimbo and Henriquez Gallegos, 2014; Ferrari et al., 2018), to the best of our knowledge, it has still not been tested in a global econometric analysis dedicated to the microfinance industry. Such a measure may however significantly affect both the MFIs and their clients. Especially, given the above arguments, caps would exacerbate financial exclusion by constraining MFIs to target better-off or more financially included clients, and as a result to provide larger loans. Therefore, we highlight the following initial hypothesis:

H1: MFIs facing interest rate caps provide larger loans on average

Many governments are not satisfied with this rationale and still implement interest rate restrictions (Christen and Rosenberg, 2000). As suggested previously, the main arguments raised are that caps would force MFIs to reduce their costs, improve efficiency, and protect the poor from being financially exploited through excessive pricing (Mitra, 2009). As explained by Gul et al. (2017:2): “both regulation and supervision by government may exert meaningful effects on MFI performance, and in either case, policy actions are likely to be motivated by government ideology”. In other words, caps may also be implemented with political motivations (Miller, 2013). In developed countries, usury limits are also used to a non-negligible extent (Maimbo and Henriquez Gallegos, 2014), although they do not generate

particularly more enthusiasm in the financial literature (Goudzwaard, 1968; Villegas, 1989; Benmelech and Moskowitz, 2010; Zinman, 2010; Rigbi, 2013).

Competition and microfinance

Beside regulation, competition is another important force shaping microfinance but it has not received a similar attention from scholars (Kar and Swain, 2018). In few decades, the sector has commercialized, bringing in formalization, the quest for sustainability, and more and more competition. Just like for traditional financial institutions (Claessens and Laeven, 2005), it has often been argued that competitive pressure allows MFIs to reduce prices, stimulate innovation and enhance product quality and diversity (Rhyne and Christen, 1999). Overall, competition has indeed contributed to reduce lending rates and to diversify microfinance services (Christen et al., 2012; Al-Azzam, 2019). Still, Porteous (2006) brings nuance and explains that competition has different effects depending on the maturity of the market considered. As an example, Porteous compares Uganda, Bangladesh and Bolivia, three intensively competitive microfinance markets in the mid-2000s. While competition reduced significantly interest rates in Bolivia (and to a lesser extent in Uganda), it has not done so in Bangladesh, where it rather materialized through the diversification of products and the client targeting.

As the sector commercialized, researchers have however also shown that competition may be detrimental when it comes to MFIs' outreach (Assefa et al., 2013). In particular, scholars point different drawbacks.

A first drawback is to be found in the deterioration of one of the most important mechanisms for MFIs to sustainably reach the poor: *cross-subsidization*. Cross-subsidizing consists in providing larger loans to better-off clients in order to deliver and compensate for smaller, costlier loans offered to poorer beneficiaries (Armendáriz and Szafarz, 2011). Yet, as a profitable business, microfinance has attracted more and more profit-oriented players. These profit-seeking actors entering the market jeopardize the possibility to cross-subsidize by reducing the basket of available better-off clients (Morduch, 1999b; McIntosh and Wydick, 2005). At the end, this prevents small, socially oriented MFIs to take advantage from cross-subsidization to serve the poorest clients requiring the tiniest loans. According to Kar and Swain (2014), this has also been exacerbated as lending rates dropped due to competition.

Second, competition may also exacerbate information asymmetry (McIntosh and Wydick, 2005). In the absence of effective information sharing platform, the information about clients' indebtedness cannot be properly checked. Poor borrowers may thus receive inadequate offers, resulting at the end of the day in lower repayment rates. Navajas et al. (2003) and McIntosh et al. (2005) respectively show through Bolivian and Ugandan data that competitive pressure drives MFIs to loosen their terms and conditions. Especially, when markets become more and more saturated, MFIs may have no option but to relax lending standards and screening mechanisms in order to keep their customers while reducing costs (Schicks and Rosenberg, 2011). This induces in some cases risky refinancing practices leading to the provision of larger loans to clients who do not have adapted repayment capacities, which strengthens the risk of over-indebtedness and affects repayment performance.

Third, MFIs operating in the most competitive markets may also struggle to reach sustainability (Vanroose and D’Espallier, 2013; Kar and Swain, 2014). As competition encourages efficiency, it makes it more difficult for small actors with less possibilities of scale economies to survive. In financial inclusion schemes, these small operators are however often NGOs, generally more socially oriented and working with poorer or less accessible clients. By pushing these smaller actors out of the market, competition may contribute to exacerbate financial exclusion.

The arguments presented above indicate that competition may, through various mechanisms, jeopardize MFIs’ outreach and deteriorate financial inclusion. Coming back to regulation and interest rate caps, MFIs may thus face extra pressure on their sustainability because of competition than what a cap already implies. Consequently, imposing such a measure in a competitive environment may exacerbate even more the financial exclusion effect of a cap suggested by the literature and detailed earlier. Therefore, we suggest the following hypothesis:

H2a: competition amplifies the effect of interest rate caps on the average loan size

Still, the opposite can also be discussed. Vogelgesang (2003) shows that clients who do not take multiple loans while facing an abundant supply of microcredit and an intense competition¹ present better repayment. According to the author, the explanation may be two-fold. Either clients in competitive markets are more aware of the danger of multiple borrowing, and/or MFIs facing intense competition pay more attention to repayments and, for instance, raise more awareness and/or provide clients with more incentives to repay. This way, risky behaviors such as multiple borrowing, unreasonable refinancing or loan toping-up practices, which tend to increase the average loan size, may not be that much adopted.

More recently, Vanroose and D’Espallier (2013) have also shown that competition from purely commercial actors encourages more socially oriented MFIs to reach niche markets, especially poorer or less accessible segments. Studying the outreach of MFIs through the average loan size they provide, Cull et al. (2014) confirm that competition from commercial banks encourage MFIs to target poorer clients and consequently to provide smaller loans.

Besides, Karimu et al. (2019) find that competition encourages MFIs to adopt riskier credit strategy. For some MFIs, additional risk may go with looking for poorer or more excluded clients. Indeed, while such clients require smaller loans, they generally lack financial literacy, identification pieces and possibilities to provide collaterals (Morduch, 1999a).

As suggested above, we could thus argue that competition may also encourage the financial inclusion of the poorest and, thereby, the provision of smaller loans. Consequently, we highlight a third hypothesis:

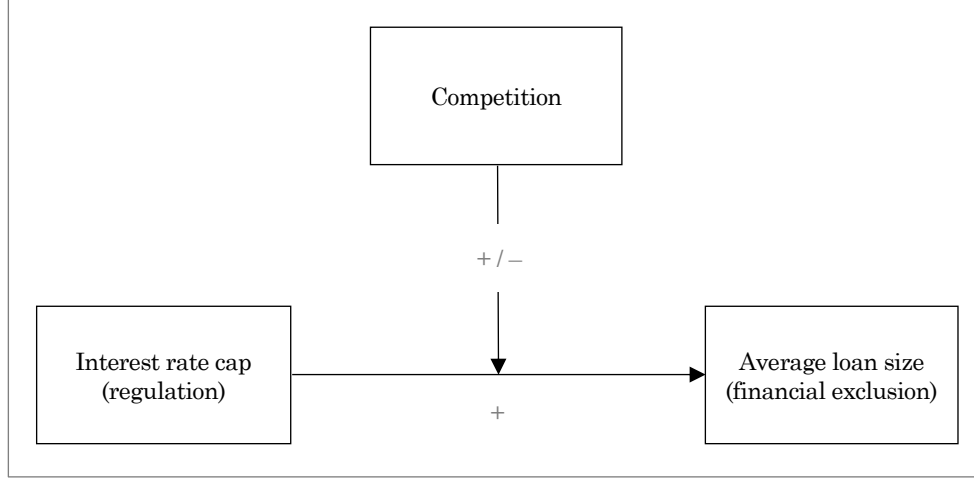
H2b: competition attenuates the effect of interest rate caps on the average loan size

As suggested by the last two hypotheses, the outcome of an interest rate cap in terms of financial inclusion may depend on market conditions, and particularly on the competition that MFIs face. MFIs may indeed not respond to an interest rate cap the same way when facing

¹ This situation is not so uncommon, see for instance MIMOSA (2016) on Cambodia.

numerous competitors than when being isolated. Hence, validating one of these last two hypotheses boils down to identifying an interaction between interest rate caps – and so a fortiori regulation – and competition, as described by Figure 1.

Figure 1. Interaction of competition with the interest rate cap – average loan size relationship



The ambiguous implications highlighted by the literature when it comes to the outcomes of competition in a financial inclusion context make it necessary to adopt a different and more contingent approach when investigating regulation and market dynamics issues. To date, and to the best of our knowledge, the recent study by Karimu et al. (2019) is the only research analyzing a combined effect of regulation and competition in microfinance. With Sub Sahara African data, the authors show that regulation reduces credit risk when competition is low but does not in more competitive markets. In this latter case, intense competition would prevent regulation from helping MFIs to adopt reasonable risk-taking strategies. The authors build an interaction variable combining a binary variable indicating whether an MFI is regulated and a measure of the competition that this MFI faces. Yet, apprehending the conditional effect that regulation has depending on competition may require a more precise variable. In fact, the core principle of regulating microfinance may be summarized as follows: MFIs collecting deposits follow prudential regulation while credit-only institutions are subject to non-prudential regulation. Hence, all (formal, legally established) MFIs generally face at least some kind of non-prudential regulation. If implemented, interest rate caps are thus likely to apply to all formal MFIs, regardless of whether they follow prudential regulation. Characterizing them as not regulated if they follow non-prudential regulation may thus be too wide to understand the outcomes of a precise regulatory measure. Therefore, using a more specific variable is necessary. This is what our paper does by analyzing the combined effect of interest rate caps and competition.

3 Empirical approach

We performed a Breusch-Pagan test and a joint F-test, which revealed the presence of individual and time effects, indicating that panel data estimations are best fitted in this case. Additionally, we suspected unobserved time-invariant effects, especially as microfinance practices and performances may be different across regions (Vanroose, 2008). Running a

Hausman Test (Hausman, 1978), we failed to reject the null hypothesis of absence of time-invariant effects, which indicates that the fixed-effects approach is best fitted.

In order to test the above hypotheses, we use the following model:

$$ALS_{i,t} = \beta_0 + \beta_1 RATECAP_{ij,t} + \beta_2 COMP_{i,t} + \beta_3 RATECAP_{ij,t} * COMP_{i,t} + \lambda X_{i,t} + \gamma Z_{j,t} + \eta_i + \mu_t + \varepsilon_{i,t} \quad (1)$$

where the dependent variable is $ALS_{i,t}$, the average loan size provided by the MFI i in year t . As a measure of social performance, the average loan size is probably the most used proxy for the depth of outreach of an MFI (Cull et al., 2007; Mersland and Strøm, 2010; Kar and Swain, 2014; 2018; Reichert, 2018). The depth of outreach can roughly be understood as the “level of poverty” of the clients reached. As explained by Schreiner (2001:7), “poorer borrowers have fewer complementary assets to combine in production with the large, lumpy assets that might be purchased with a large disbursement. For example, a farmer with two hectares of land is unlikely to use a disbursement to buy a tractor.” (Schreiner, 2001:7). Therefore, the smaller the loan, the poorer the client is supposed to be (Schreiner, 2002; Olivares-Polanco, 2005). To manipulate comparable measures across countries, we divide ALS by the GDP per capita (Cull et al., 2007).

Our first explanatory variable of interest is $RATECAP_{ij,t}$, a dummy variable indicating whether there is a cap in the country j in year t . As this paper focuses on the way regulation affects financial inclusion, we only consider “hard” interest rate caps, and not “soft” ones (Zetzsche and Dewi, 2018). Hence, caps resulting from public or political pressures (Helms and Reille, 2004) are not considered here. $COMP_{i,t}$ is then our explanatory variable for competition, while $RATECAP_{ij,t} * COMP_{i,t}$ is an interaction variable capturing the combined effect of the imposition of an interest rate cap and the level of competition. We first measure competition based on a “Structural” or market-based approach (Kar and Swain, 2018). The structural approach considers market structure to apprehend the level of competition. Especially, it analyses the number of organizations competing in the industry – and thereby market concentration – as a proxy for market power or competition. In this perspective, the traditional HHI is probably one of the most used indicators. Compared to other concentration ratios, the HHI integrates all the firms of an industry, and not only the larger ones, and allows to better consider the size of the firm, as it penalizes more large market shares through square weights (Purkayastha et al., 2017). Just like in many fields the HHI has largely been used in the microfinance literature (Wagner and Winkler, 2013; Kar and Swain, 2014; Ayele, 2015; Mia, 2018; Al-Azzam and Parmeter, 2019; Dannon et al., 2019). The HHI is calculated as:

$$HHI_{ij,t} = \sum_{i=1}^n MS_{i,t}^2 \quad (2)$$

where $MS_{i,t}$ is the market share of the MFI i (based on the gross loan portfolio) in year t . We compute the HHI for each country and each year to account for cross-sectional as well as time variations (Kar and Swain, 2014). The HHI takes values between 0 (perfect competition) and 1 (perfect monopoly). Therefore, the higher the indicator, the lower the competition. For ease of interpretation, we multiply the HHI by -1 so that an increase in HHI indicates an increase of competition. As detailed in Section 5, we also use the number of firms in the industry and the Lerner Index as additional robustness checks to apprehend competition.

Our model also includes a bunch of common control variables to capture both firm-level and country-level influences. $X_{i,t}$ is a vector of MFI-specific variables including the percentage

of female (*WOMEN*) and rural (*RURAL*) borrowers served, the total assets (*ASSETS*), the number of borrowers served (*BREADTH*), the collection of savings (*SAVINGS*, binary variable), a dummy variable indicating whether an MFI is sustainable (*SUSTAINABLE*), the operating expense ratio (*OER*) and the *PAR30* ratio. Furthermore, $Z_{j,t}$ is a vector of country-specific variables characterizing the economic environment, including the GDP growth rate (*GROWTH*), the inflation rate (*INFLATION*), the percentage of rural population (*RURALPOP*), the manufacturing value added in percentage of GDP (*MANUFACT*), and the credit to GDP ratio (*CREDIT*). All above variables have been used in major cross-country studies analysing the determinants of the average loan size or using it as a proxy for depth of outreach (Conning, 1999; Olivares-Polanco, 2005; Gonzalez and Rosenberg, 2006; Cull et al., 2007; Hartarska and Nadolnyak, 2007; Vanroose, 2008; Mersland and Strøm, 2010; Ahlin et al., 2011; Assefa et al., 2013; Vanroose and D’Espallier, 2013). In addition, we include the share of the portfolio dedicated to SMEs (*SME*), as SMEs certainly receive larger loans than microborrowers. We also include the ratio of donated equity on total liabilities (*DONATED*) to indicate the extent to which an MFI relies on donations. As we lack information on the legal form of MFIs as such, this is also a way to capture the influence of the orientation of the MFIs that may eventually affect outreach and performance (Cull et al., 2009; Roberts, 2013). Appendix 1 provides a description of the variables. Finally, we integrate time dummies to account for time-specific effects. $\varepsilon_{i,t}$ is the error term.

4 Data and descriptive statistics

We built a consolidated dataset comprising 1124 MFIs over 2015-2018 and out of 83 countries, representing a total of 3156 observations. It is to note that punctual errors were noticed in the dataset. When the error could be corrected, for instance by cross-checking with the dataset in local currencies, we corrected the observations. Yet, we excluded the observations for which ALS was in the first and last percentiles, as some of the errors were not rectifiable. We also removed the observations for which the HHI was equal to 1, as they reflected a clear lack of data for a specific country, rather than a perfectly monopolistic market. Due to these adjustments, the final dataset comprised 986 MFIs, out of 73 countries, representing a total of 2591 observations.

Data being required at multiple levels, we built this dataset based on several sources. First, we used MFI-level data from the Microfinance Information eXchange (MIX Market). The MIX is still to date the largest effort in the microfinance industry to collect data around the world and has been used in major studies related to microfinance and financial inclusion (Cull et al., 2007, 2009; Hartarska and Nadolnyak, 2007; Ahlin et al., 2011; Assefa et al., 2013; Bauchet and Morduch, 2013; D’Espallier et al., 2013; Gul et al., 2017). Although the platform shut down its activities in late 2019, the data collected are now publicly available and hosted by the World Bank. Also, since some MFIs started-up while others bankrupted in the period of interest, the dataset consists in an unbalanced panel. Second, regarding the use of interest rate caps around the world, precise information is a priori not centralized nor uniformly classified (Ferrari et al., 2018). We thus gathered data through different channels. We started by looking at the *Global Microscope on Financial Inclusion* (GMFI), an initiative from The Economist Group carried out every year since 2007 and which analyzes the regulatory environment where MFIs operate in about fifty countries. Among other things, the GMFI

informs on whether the countries impose interest rate caps. However, to double-check as well as to complement this, we used the studies by Maimbo and Henriquez Gallegos (2014), Sinha (2016), Ferrari et al. (2018) and Zetzsche and Dewi (2018), which provide useful inventories of the use of interest rate caps in multiple countries. In addition, we carried out a desk review of related national legal documents as well as the websites of local central banks, press agencies and even had direct contacts with central bankers, networks, and other field actors. When we had too few or contradictory information about whether there are interest rate caps in a given country, we did not take this particular country into account. Although we collected data for 101 countries, only 73 remained after consolidating the dataset for the countries included in the MIX market for 2015-2018 and adjusting the data for the reasons explained earlier. Third, we used the World Development Indicators (WDI) for broad economic data.

Overall, 31 countries capped interest rates for the whole period, while 35 did not. Yet, interest rate caps are not necessarily static (Ferrari et al., 2018), and 6 countries implemented a cap in the period, while 1 removed it. Appendix 2 provides details all these countries.

Furthermore, Table 2 shows the results of a mean-comparison test (t-test) investigating the difference in average loan sizes between countries where rates are capped compared to those where they are freely determined. Prior to regressions, a significant difference is observed with a confidence level of 95%, suggesting, as hypothesized, that MFIs located in countries capping interest rates offer larger loans on average². As shown in Table 3, the average HHI, our main proxy for competition, is 0.277, which suggests a moderate competition in the microfinance sector globally speaking. All in all, the mean values given by Table 3 for the (adjusted) average loan size and the HHI are similar to those from other studies (Cull et al., 2007; Assefa et al., 2013; Kar and Swain, 2014; Dannon et al., 2019). Eventually, Table 4 presents the correlations.

Table 2. Mean-comparison test (t-test) for the average loan size, by state of RATECAP

	N	Mean	SD	Min	Max
<i>No cap</i>	1065	1332.619	2521.694	43	32613
<i>Cap</i>	1387	1778.421	2657.336	71	48417
<i>Difference</i>		(-445.802)*			
<i>Combined</i>	2452	1584.792	2608.149	43	48417

* significance level of 95%

² Identical results are observed when running the test with ALS adjusted by the GDP per capita.

Table 3. Summary statistics

Variable	N	Mean	SD	Min	Max
<i>Dependent variables</i>					
<i>Average loan size adj. GDP per capita</i>	2591	0.537	0.772	0.023	5.713
<i>Explanatory variables</i>					
<i>Interest rate cap (dummy)</i>	2452	0.566	0.496	0	1
<i>Herfindahl-Hirschman Index</i>	2590	0.277	0.212	0.065	0.999
<i>Percentage of female borrowers</i>	2171	0.644	0.263	0	1
<i>Percentage of rural borrowers</i>	1972	0.525	0.323	0	1
<i>Share of portfolio dedicated to SMEs</i>	1836	0.133	0.224	0	1
<i>Total assets (log)</i>	2468	16.765	2.033	10.775	22.786
<i>Number of active borrowers (log)</i>	2591	9.867	2.082	2.565	15.741
<i>Savings (dummy)</i>	2591	0.500	0.500	0	1
<i>Donated equity on total liabilities</i>	1981	0.040	0.134	-0.187	1.585
<i>Sustainable (dummy)</i>	2591	0.826	0.379	0	1
<i>Operating Expense ratio</i>	2230	0.225	0.225	0.0002	4.927
<i>PAR30</i>	2397	0.070	0.117	0	12159
<i>GDP growth rate</i>	2591	0.042	0.028	-0.059	0.104
<i>Inflation rate</i>	2402	0.040	0.036	-0.024	0.295
<i>Share of rural population</i>	2573	0.480	0.201	0.081	0.879
<i>Manufacturing value added (in % of GDP)</i>	2561	0.135	0.041	0.017	0.295
<i>Credit to GDP ratio</i>	2531	0.425	0.231	0.055	1.611

Table 4. Correlation matrix

	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]	[14]	[15]	[16]	[17]	[18]
[1] <i>ALS adj. GDP p.c.</i>	1.0000																	
[2] <i>RATECAP</i>	0.0525*	1.0000																
[3] <i>HHI</i>	0.1200*	-0.0590*	1.0000															
[4] <i>WOMEN</i>	-0.4036*	0.0668*	-0.1815*	1.0000														
[5] <i>RURAL</i>	-0.1489*	0.0034	-0.0706*	0.1601*	1.0000													
[6] <i>SME</i>	0.2442*	-0.1448*	0.0799*	-0.1226*	-0.0108	1.0000												
[7] <i>ASSETS</i>	0.2424*	0.0672*	-0.1066*	-0.0884*	-0.0678*	0.0781*	1.0000											
[8] <i>BREADTH</i>	-0.1197*	0.0569*	-0.1512*	0.3391*	0.1219*	-0.0282	0.8029*	1.0000										
[9] <i>SAVINGS</i>	0.2676*	-0.0234	0.1331*	-0.0736*	-0.0070	0.1063*	0.3111*	0.2158*	1.0000									
[10] <i>DONATED</i>	-0.0789*	-0.0082	0.0925*	-0.0211	0.0994*	-0.0418	-0.2745*	-0.2508*	-0.1714*	1.0000								
[11] <i>SUSTAINABLE</i>	-0.0642*	0.0024	-0.0684*	0.1414*	0.0281	-0.0857*	0.1586*	0.1631*	-0.0100	-0.1067*	1.0000							
[12] <i>OER</i>	-0.1513*	-0.2628*	0.1113*	0.0240	-0.1636*	0.0040	-0.2573*	-0.2086*	-0.0351	0.0421	-0.2426*	1.0000						
[13] <i>PAR30</i>	0.1131*	-0.0034	0.0203	-0.2247*	-0.1069*	0.1574*	0.1214*	-0.1942*	-0.0132	0.0234	-0.2539*	0.0730*	1.0000					
[14] <i>GROWTH</i>	0.0479*	0.0055	-0.0743*	0.3820*	0.2792*	0.1128*	-0.0148*	0.2508*	0.1022*	-0.0497*	0.0768*	-0.0897*	-0.1529*	1.0000				
[15] <i>INFLATION</i>	-0.0783*	-0.0353	0.2099*	0.0534*	0.0069	0.0984*	-0.0676*	0.0267	-0.0566*	0.0309	-0.0207	0.0516*	0.0995*	-0.2147*	1.0000			
[16] <i>RURALPOP</i>	0.2066*	0.1199*	0.0335	0.2629*	0.3723*	0.1120*	0.0510*	0.1748*	0.2406*	0.0011	-0.0574*	-0.1721*	-0.0299	0.6112*	0.0555*	1.0000		
[17] <i>MANUFACT</i>	-0.2073*	0.0151	-0.3525*	0.3238*	0.1034*	-0.0585*	-0.0878*	0.1827*	-0.0704*	-0.0917*	0.1302*	0.0400	-0.1397*	0.1868*	-0.1942*	-0.1413*	1.0000	
[18] <i>CREDIT</i>	-0.0448*	0.1457*	0.0956	0.1517*	0.0696*	-0.0882*	-0.1202*	0.0648*	-0.1443*	0.0508*	0.0963*	-0.1926*	-0.1152*	0.1139*	-0.1531*	-0.0838*	0.2035*	1.0000

* significance level of 95%.

5 Results

Our analysis aims at identifying the cross-country effect of interest rate caps on the average loan size adjusted by the GDP per capita, and understanding whether the intensity of competition moderates this effect. Table 5 presents the results of the estimations³.

Looking first at Models 1 to 3 in Table 5, we observe a positive and significant coefficient for *RATECAP*. Everything being equal, it means that MFIs operating in markets where regulators implement interest rate caps tend to provide larger loans on average to their customers, which validates our first hypothesis. Due to microfinance’s business model, relatively high operational costs are inevitable to provide small loans, when compared to traditional financial institutions. Moreover, the poorest clients generate additional costs, especially because of their remoteness or their need for complementary (and costly) non-financial services. Therefore, this first result suggests that interest rate caps, by constraining MFIs to lower their interest rates and reduce their margins, inevitably reduce the profit they generate. In order to maintain a steady level of profit or to keep breaking-even, MFIs increase the size of the loans they provide and target clients who are less costly to serve. As the average loan size proxies the level of poverty of the clients reached, this finding suggests that MFIs facing interest rate caps reach out relatively poorer clients than MFIs operating with freely determined rates, and that capping interest rates in microfinance exacerbates the financial exclusion of the poorest. This finding corroborates what has long been argued in both the literature and the field but so far not tested in a global quantitative study on microfinance markets.

Model 4 then integrates our competition variable, *HHI*. The positive and highly significant coefficient of *HHI* indicates that, everything else being equal, the higher *HHI*, the larger *ALS*. From a market-level perspective, this suggests that when a market gets less concentrated – or more competitive – MFIs tend to provide larger loans on average. This finding goes in the sense of previous research showing that competition can damage microfinance outreach or social performance generally speaking (Navajas et al., 2003; Vogelgesang, 2003; McIntosh and Wydick, 2005; McIntosh et al., 2005; Assefa et al., 2013; Kar and Swain, 2018; and Karimu et al., 2019).

Furthermore, Model 5 includes the variables resulting from the interaction of *HHI* and *RATECAP*. The significant coefficient of *RATECAP* shows a positive direct effect. This means that in markets where *HHI* equals 0, the MFIs facing an interest rate cap tend to provide larger loans to their clients. Although it goes in the sense of the results highlighted above, this coefficient should be cautiously considered, as a value of 0 for *HHI* implies a perfectly competitive market. While it fully makes sense mathematically speaking, such market does not exist (neither in the data, nor in reality). Besides, the positive and significant coefficient of *HHI* indicates a positive direct effect of competition. That is, as competition increases, the average loan size increases too, conditionally to the fact that there is no cap on interest rates.

³ The estimates presented in Table 5 are robust to heteroscedasticity and serial correlation. We also performed a Durbin-Wu-Hausman endogeneity test (augmented regression test) as proposed by Davidson-MacKinnon (1993), which indicated no simultaneity issue and whose results are available on request.

Looking now at the interaction variable, the positive and significant coefficient suggests that, as competition changes, the effect of *RATECAP* on *ALS* increases. In other words, competition amplifies the initial relationship observed between *RATECAP* and *ALS*, which validates hypothesis H2a. This positive interaction relationship can be visualized on Figure 2, especially through the increase in slope illustrated by the dash line.

Figure 2. Effect of *RATECAP* on *ALS* for different values of *HHI*

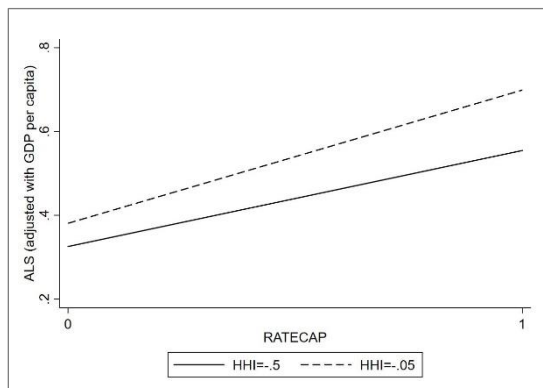
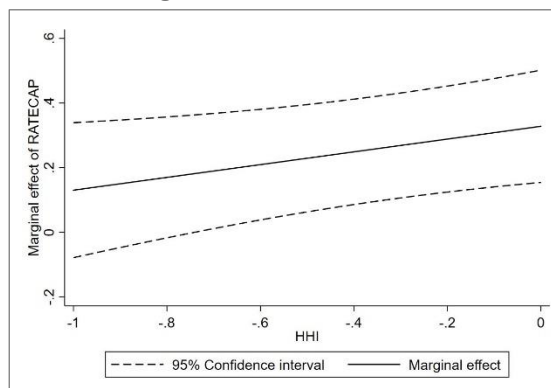


Figure 3. Marginal effect of *RATECAP* on *ALS* as *HHI* changes



These results may find a first explanation by coming back to the process of cross-subsidization detailed earlier. When there is more competition, the basket of better-off clients is split into more MFIs, this way reducing their possibility to cross-subsidize their loans and forcing them to finance fewer poor people. In a situation where their interest rates are legally limited, competition just adds to their burden by limiting a bit more the number of wealthier clients thanks to which MFIs could earn profit and offer smaller loans. Another explanation may also lie with the increasing need for more efficiency induced by competition. Small operators, generally NGOs, are often less efficient. Nevertheless, these small, generally more socially oriented, entities also tend to serve poorer clients. In such a context, the harsher competition may push these operators out of the market, this way increasing the overall average loan size and exacerbating even more financial exclusion. As explained by Helms and Reille (2004) and Acclassato (2008), interest rate caps exacerbate this phenomenon.

Yet, so far, nothing tells that this moderation effect manifests for all states of competition. It is thus useful to “probe” this effect (Hayes, 2018), for all the values of the moderator, *HHI*. One way to do so is to analyze the conditional effect of *RATECAP* at different values of *HHI* and check for which values it is significant. This is what Figures 3 does. The confidence interval comprises 0 for values of *HHI* from -1 to -0.83. This would mean that, for these values, the moderation effect is not significant. However, the cumulative frequency of the distribution of *HHI* in our dataset shows that it only concerns 2.20% of our observations. MIX Market being based on self-reported data, the observations included in this range certainly suffer from a lack of data collection in the related countries, pushing upwards the observed values of *HHI*, rather than reflecting truly quasi-perfect monopolistic markets. Furthermore, we can observe that the moderation effect of competition is significant for the other 97.80% observations. All in all, our results suggest that the moderation effect of competition exists for almost the whole spectrum of *HHI*, that is, for most microfinance industries.

Table 5. Estimations for the average loan size adjusted with GDP per capita (fixed-effects regressions)

ALS (adj. GDP p.c.)	(1)	(2)	(3)	(4)	(5)
<i>RATECAP</i>	0.198*** (0.053)	0.355*** (0.099)	0.280*** (0.083)	0.278*** (0.082)	0.328*** (0.088)
<i>HHI</i>				0.243*** (0.063)	0.123* (0.074)
<i>RATECAP * HHI</i>					0.197** (0.098)
<i>MFI-level controls</i>					
<i>WOMEN</i>		-0.095 (0.061)	-0.051 (0.049)	-0.051 (0.050)	-0.055 (0.052)
<i>RURAL</i>		0.036 (0.035)	0.034 (0.040)	0.035 (0.040)	0.036 (0.041)
<i>SME</i>		-0.009 (0.056)	-0.020 (0.052)	0.004 (0.046)	0.002 (0.045)
<i>ASSETS</i>		0.268*** (0.060)	0.198*** (0.064)	0.197*** (0.064)	0.197*** (0.064)
<i>BREADTH</i>		-0.358*** (0.063)	-0.305*** (0.063)	-0.299*** (0.062)	-0.299*** (0.0621)
<i>SAVINGS</i>		-0.075 (0.050)	-0.063 (0.052)	-0.047 (0.054)	-0.043 (0.055)
<i>DONATED</i>		-0.001 (0.140)	0.043 (0.122)	0.064 (0.101)	0.064 (0.096)
<i>SUSTAINABLE</i>		0.018 (0.022)	0.016 (0.019)	0.024 (0.019)	0.025 (0.018)
<i>OER</i>		-0.133 (0.117)	-0.116 (0.108)	0.115 (0.107)	-0.117 (0.106)
<i>PAR30</i>		0.082 (0.087)	0.038 (0.076)	0.044 (0.078)	0.048 (0.080)
<i>Country-level controls</i>					
<i>GROWTH</i>	-0.012 (0.274)		-0.368 (0.426)	-0.492 (0.419)	-0.517 (0.422)
<i>INFLATION</i>	-0.186 (0.199)		0.370 (0.310)	0.696** (0.341)	0.734** (0.347)
<i>RURALPOP</i>	-2.670*** (0.858)		2.021 (1.829)	3.639* (2.001)	2.917 (1.886)
<i>MANUFACT</i>	-1.182 (1.148)		0.090 (1.140)	-0.180 (1.107)	0.040 (1.082)
<i>CREDIT</i>	0.233** (0.094)		0.034 (0.143)	-0.100 (0.143)	-0.176 (0.154)
<i>Time dummies</i>	No	No	Yes	Yes	Yes
<i>_cons</i>	1.685*** (0.446)	-0.543 (0.670)	-0.866 (1.127)	-1.495 (1.159)	-1.196 (1.116)
<i>Overall R squared</i>	0.022	0.1937	0.2617	0.1836	0.219
<i>N</i>	2243	1229	1132	1132	1132
<i>Number of MFIs</i>	893	630	596	596	596

Notes: standard errors in parentheses; significance levels as *** (1%), ** (5%) and * (10%); “Yes” indicates that time dummies are included and significant at 10% or better, “No” indicates that time dummies are not included.

Robustness checks

Although important, market concentration may not necessarily be the only factor influencing competition. Especially, market concentration does not account for geographical differences between countries (Van Leuvensteijn et al., 2007). The HHI may thus not perfectly capture the effect of competition. Therefore, we also used other indicators as robustness checks, namely the number of firms in the industry and the Lerner Index (LI)⁴. While the former somewhat refers to the same rationale as the HHI, the latter relies on a “Non-Structural” or organization-based approach consisting in observing the conduct of the firms as an indication of the competition they face (Kar and Swain, 2018). More concretely, this approach considers prices, costs, margins or elasticities as reflecting the competitive pressure. The LI has frequently been used as a firm-level measure of competition in the financial as well as microfinance literature (Maudo and Fernández de Guevara, 2004; Koetter et al., 2012; Assefa et al., 2013; Purkayastha et al., 2017; Al-Azzam and Parmeter, 2019; Dannon et al., 2019). The LI is computed as follows⁵:

$$LI_{i,t} = \frac{(P_{i,t} - MC_{i,t})}{P_{i,t}} \quad (3)$$

where $P_{i,t}$ is the price of the output, defined here as the ratio of financial revenues from loans on the gross loan portfolio (Al-Azzam and Parmeter, 2019), and $MC_{i,t}$ is the marginal cost. To compute MC , we first estimated the following translog cost function:

$$\begin{aligned} \ln C_{i,t} = & \alpha_0 + \alpha_1 \ln Q_{i,t} + \frac{1}{2} \alpha_2 (\ln Q_{i,t})^2 + \sum_{j=1}^3 \beta_j \ln W_{j,i,t} + \frac{1}{2} \sum_{j=1}^3 \beta_j (\ln W_{j,i,t})^2 + \sum_{j=1}^3 \gamma_j \ln Q_{i,t} * \ln W_{j,i,t} + \\ & \sum_{j < k} \sum \gamma_{j,k} \ln W_{j,i,t} * \ln W_{k,i,t} + \delta_1 TREND + \frac{1}{2} \delta_2 TREND^2 + \delta_3 \ln Q_{i,t} * TREND + \sum_{j=1}^3 \eta_j \ln W_{j,i,t} * \\ & TREND + \zeta_1 \ln PAR30_{i,t} + \zeta_2 \ln ASSETS_{i,t} + \zeta_3 SAVINGS_{i,t} + \zeta_4 COUNTRY + \varepsilon_{i,t} \end{aligned} \quad (4)$$

where $C_{i,t}$ is the total cost, the ratio of the total expenditures on the gross loan portfolio. $Q_{i,t}$ is then the output, namely the gross loan portfolio. We consider the cost of three inputs ($W_{j,i,t}$) including the cost of labor (personnel expenses on gross loan portfolio), the cost of physical capital (administrative expenses on net fixed assets), and the cost of financial capital (financial expenses on liabilities). We also include the *PAR30*, the total assets and a dummy for the collection of savings to account for differences respectively in risk-taking strategy, size and service offer across MFIs (Assefa et al., 2013; Al-Azzam and Parmeter, 2019). A time trend is also included to account for changes in performance induced, for instance, by technological change (Karimu et al., 2019). We eventually add a vector of dummies to capture potential country-specific effects influencing the cost structure (Al-Azzam and Parmeter, 2019). For better estimations, the cost function is estimated jointly with cost share equations (derivatives of the cost function with respect to each input cost) using seemingly unrelated regression, and with cross-equation parameters constraints (Kar and Swain, 2018). The estimates are then

⁴ Other measures exist to apprehend competition, just like the H-statistic of the Panzar-Rosse Model or the Boone indicator. Yet, the H-statistic does not allow time variations, which is irrelevant to assess the evolution of competition (Assefa et al., 2013). Besides, the Boone indicator uses a translog cost function, just like the Lerner Index. Still, it requires enough observations to compute the elasticity of profit to marginal cost for each country-year couple. Given the number of countries included in our dataset, we favored the Lerner Index.

⁵ Although, from a theoretical perspective, the Lerner Index may be influenced by a cap on interest rates, the correlation between both variables is very weak (0.07). Additionally, the corresponding Variance Inflation Factors (VIFs) are very close to 1, suggesting no multicollinearity issue in this case (Kleinbaum et al., 1988).

used in equation 5 to compute MC , defined as the derivative of the cost function (equation 4) with respect to $\ln Q_{it}$:

$$MC_{i,t} = \left(\frac{c_{i,t}}{Q_{i,t}} \right) * \left(\alpha_1 + \alpha_2 \ln Q_{i,t} + \sum_{j=1}^3 \gamma_j \ln w_{j,i,t} + \delta_3 trend \right) \quad (5)$$

Once computed, $MC_{i,t}$ is reinjected in equation 3 to calculate the LI for each MFI and each year. Appendix 3 shows summary statistics of the additional variables used for the estimation of the cost function and the calculation of the LI. Just like the HHI, the LI ranges between 0 and 1⁶. The intuition is that firms in highly competitive markets (LI close to 0) are price takers, inducing that P is close to MC . On the contrary, oligopolies or monopolies allow for prices to be set above the marginal cost, inducing values closer to 1 (Elzinga and Mills, 2011).

Table 6 shows that both the number of MFIs ($NBMFI$) in the industry and the LI ($LERNER$) provide similar results than HHI when estimating the interaction effect of both variables with the $RATECAP$ -ALS relationship. Similar results are also observed when probing the interaction variables with a 95% confidence interval, just like for HHI .

Table 6. Estimations for the average loan size adjusted with GDP per capita (fixed-effects regressions)

ALS (adj. GDP p.c.)	Using LERNER	Using NBMFI
<i>RATECAP</i>	0.696*** (0.170)	0.197** (0.080)
<i>LERNER^a</i>	-0.609*** (0.210)	
<i>RATECAP * LERNER^a</i>	0.653*** (0.207)	
<i>NBMFI</i>		-0.001 (0.001)
<i>RATECAP * NBMFI</i>		0.006*** (0.001)
<i>Firm-level control variables</i>	Yes	Yes
<i>Country-level control variables</i>	Yes	Yes
<i>Time dummies</i>	Yes	Yes
<i>_cons</i>	-2.634*** (0.931)	-0.829 (1.126)
<i>Overall R squared</i>	0.317	0.213
<i>N</i>	857	1132
<i>Number of MFIs</i>	454	596

Notes: *LERNER* multiplied by (-1) for ease of interpretation; standard errors in parentheses; significance levels as *** (1%), ** (5%) and * (10%); "Yes" indicates that the variables are included in the model.

As additional robustness checks on the dependent variable, we ran our main model for the average loan size adjusted by the Gross National Income (GNI) per capita, as well as without any adjustment. We obtained similar results than when adjusting the average loan size with the GDP per capita.

⁶ We still observed 17 negative observations. Although it is possible from a theoretical perspective, we considered these as outliers, as a negative value can be observed only if MC is greater than the marginal return, which would indicate that an MFI is heavily subsidized to be able to keep running (Assefa et al., 2013). We ran our models while computing the Lerner Index with and without these negative values, and this did not alter the results. Therefore, we removed these observations from the dataset to remain with values comprised between 0 and 1.

6 Conclusion

Two essential trends have shaped the microfinance industry over the years: commercialization, accompanied by the exacerbation of competition, and the intensification of regulation. As microfinance has matured in multiple regions, questioning which regulation and market dynamics are desirable to consolidate the industry is now key.

On the one hand, the literature has attributed ambiguous effects to competition. While it is supposed to reduce costs and prices and encourage innovation, competition may also pressure MFIs to be more and more profitable, to the detriment of their social mission. As said Bylander et al. (2019:149) say: “as [microfinance] has become more commercial it has become more prone to crisis”. On the other hand, regulating microfinance is today considered as essential, but maladapted regulation can also destabilize the industry. Especially, interest rate caps have generated a certain amount of debates in the literature. Yet, assessing the outcome of such measures in microfinance has remained difficult, especially because of a lack of centralized data. In addition, the combined effect of interest rate caps and competition has not been considered in previous research, while the impact of such a regulation is likely to be contingent to market conditions.

In such a context, the aim of this paper was two-fold. First, thanks to substantial data that we collected on the use of interest rate caps around the world, we assessed the impact of such measures on financial inclusion. Our results show that institutions facing interest rate caps provide larger loans on average, suggesting that they tend to serve better-off or less costly clients. This confirms the idea that interest rate caps exacerbate financial exclusion. Going one step further, we carried out a moderation analysis to understand better how competition affects this initial relationship. Using multiple indicators, our results show that competitive pressure tends to amplify the phenomenon, and that this effect is valid for most states of competition. Doing so, we bring two major contributions. First, studies on interest rate caps and microfinance are generally either conceptual, centred on a specific case study, or constituted of inventories or taxonomies of caps around the world. By assessing the effect of caps on financial inclusion from a more quantitative and global perspective, this paper fills a gap in the microfinance literature. Second, almost no research studies the combined effect of regulation and competition. Yet, market conditions – and a fortiori the degree of competition – are essential to consider to understand how regulation shapes MFIs’ behavior and affects poor clients. Our research shows the importance of adopting a more systemic approach to study the outcomes of regulatory choices, overall when considering measures addressing the fate of marginalized and financially distressed individuals.

Further research may refine the analysis by bridging some of the gaps of our study. First, we only considered “hard” interest rate caps. Further investigations would thus be interesting to extend the analysis to the effects of “soft” restrictions. Second, poverty is a multi-faceted concept. Although the average loan size is a widely used proxy, further analysis may be useful to observe whether other indicators or dimensions of poverty and financial inclusion confirm the results presented in this research. Third, competition in the microfinance industry comes nowadays from a plethora of actors, including not only MFIs but also informal lenders, commercial banks, and, more recently, digital financial operators such as Mobile Network Operators or Fintechs. The nature and intensity of these different types of competition may

generate different outcomes for the MFIs and their clients. Although some authors approached this issue (Vanroose and D'Espallier, 2013; Cull et al., 2014), it is still understudied. Therefore, further research may seek to understand whether and how the competition of these other or new actors affect the behavior of traditional MFIs, and how regulators can better understand market conditions in order to integrate these actors and optimize regulatory decisions.

References

- Acclassato, D. H. (2008), "Les plafonnements de taux d'intérêt en microfinance servent-ils réellement les pauvres et petits opérateurs économiques ? ", *Mondes en Développement*, 1(141): 93-109.
- Ahlin C., Lin J. and Maio M. (2011), "Where does microfinance flourish? Microfinance institution performance in macroeconomic context", *Journal of Development Economics*, 95, 105-120.
- Al-Azzam and Parmeter (2019), "Competition and microcredit interest rates: international evidence", *Empirical Economics*.
- Armendáriz, B. and Szafarz A., (2011), "On mission drift in microfinance institutions", in *The Handbook of Microfinance*, 341-366, Armendáriz B. and Labie M., Singapore: World Scientific.
- Arun T. (2005), "Regulating for development: the case of microfinance", *The Quarterly Review of Economics and Finance*, 45: 346-357.
- Assefa E., Hermes N. and Meesters A. (2013), "Competition and the performance of microfinance institutions", *Applied Financial Economics*, 23(9): 767-782.
- Attuel-Mendès, L. and Ashta A. (2008) "French legislation and the development of credit availability for microenterprise", *Global Journal of Business Research*, 2(2): 123-137.
- Ayele G.T. (2015), "Microfinance institutions in Ethiopia, Kenya and Uganda: loan outreach to the poor and the quest for financial viability", *African Development Review*, 27(2), 117-129.
- Bauchet J., Morduch J. (2013), "Is micro too small? Microcredit vs. SME finance", *World Development*, 43, 288-297.
- Benmelech, E., and Moskowitz, T. J. (2010), "The political economy of financial regulation: Evidence from US state usury laws in the 19th century", *The Journal of Finance*, 65(3): 1029-1073.
- Bylander M., Res P., Jacoby L., Bradley P., Pérez A.B. (2019), "Over-indebtedness and microcredit in Cambodia: moving beyond borrower-centric frames", *Development Policy Review*, 37, 140-160.
- Campion, A., Ekka, R. K., and Wenner, M. (2010). "Interest rates and implications for microfinance in Latin America and the Caribbean", *IDB Working Paper Series*, n°IDB-WP-177, 52 pages.
- Christen R.P. and Rosenberg R. (2000), "Regulating microfinance – the options", *Small Enterprise Development*, 11(4): 4-23.
- Christen R.P., Lauer K., Lyman T.R. and Rosenberg R. (2012), *A guide to regulation and supervision of microfinance*, CGAP, 128 pages.
- Claessens S. and Laeven L. (2005), "Financial dependence, banking sector competition, and economic growth", *Journal of European Economic Association*, 3(1): 179-207.
- Conning J. (1999), "Outreach, sustainability and leverage in monitored and peer-monitored lending", *Journal of Development Economics*, 60, 51-77.
- Cull R., Demirgüç-Kunt A. and Morduch J. (2007), "Financial performance and outreach: a global analysis of leading microbanks", *The Economic Journal*, 117: F107-F133.
- Cull R., Demirgüç-Kunt A. and Morduch J. (2009), "Microfinance meets the market", *Journal of Economic Perspectives*, 23(1), 167-192.
- Cull R., Demirgüç-Kunt A. and Morduch J. (2014), "Banks and microbanks", *Journal of Financial Services Research*, 46: 1-53.

- Dannon H., Bangaké C., Egguh J. (2019), « Structure de marché et performance économique des institutions de microfinance dans l'UEMOA : cas du Bénin et du Togo », *Mondes en Développement*, 1(185), 29-44.
- Davidson R. and MacKinnon J.G. (1993), *Estimation and inference in econometrics*, New York: Oxford University Press.
- D'Espallier B., Guérin I., Mersland R. (2013), "Focus on women in microfinance institutions", *The Journal of Development Studies*, 49(5), 589-608.
- Elzinga K. and Mills D.E. (2011), "The Lerner index of monopoly power: origins and uses", *American Economic Review: Papers and Proceedings*, 101(3), 558-564.
- Ferrari A., Masetti O. and Ren J. (2018), "Interest rate caps: the theory and practice", *Policy Research Working Paper*, N°8398, 42 pages.
- Gonzalez-Vega C. (1977), "Interest rate restrictions and income distribution", *American Journal of Agricultural Economics*, 59(5): 973-976.
- Gonzalez A. and Rosenberg R. (2006), "The state of microfinance – Outreach, profitability, and poverty", Available at <https://ssrn.com/abstract=1400253>, accessed on 26th February 2020.
- Goudzwaard, M.B. (1968), "Price ceilings and credit rationing", *The Journal of Finance*, 23(1): 177-185.
- Guérin I., Labie M. and Morvant-Roux S. (2018), "Inadequate growth, over-indebtedness, and crises in microcredit: what have we learned?", *Enterprise Development and Microfinance*, 29(2): 118-132.
- Gul F., Podder J. and Shahriar A. (2017), "Performance of microfinance institutions: does government ideology matter?", *World Development*, 100: 1-15.
- Hartarska V. and Nadolnyak D. (2007), "Do regulated microfinance institutions achieve better sustainability and outreach? Cross-country evidence", *Applied Economics*, 39:1207-1222.
- Hausman, J. A. (1978), "Specification tests in econometrics", *Econometrica*, 46(6): 1251–1271.
- Hayes A.F. (2018), *Introduction to mediation, moderation, and conditional process analysis: a regression-based approach*, 2nd ed., The Guilford Publications, New York, 692 pages.
- Helms, B. and Reille, X. (2004), "Interest Rate Ceilings and Microfinance: the Story So Far", *CGAP Occasional Papers*, N°9, 19 pages.
- Hudon, M. (2007), "Fair interest rates when lending to the poor", *Ethics and Economics*, 5(1), 1-8.
- Kar A.K. and Bali Swain R.B. (2014), "Competition in microfinance: does it affect performance, portfolio quality, and capitalization?", in *Microfinance Institutions: Financial and Social performance*, 208-226, Mersland R. and Strøm R.O., Palgrave MacMillan.
- Kar A.K. and Swain R. (2018), "Competition, performance and portfolio quality in microfinance markets", *The European Journal of Development Research*, 30, 842-870.
- Karimu A., Salia S., Hussain J.G. and Tingbani I. (2019), "Are competitive microfinance services worth regulating? Evidence from microfinance institutions in Sub-Saharan Africa", *International Journal of Financial Economics*, 1-17.
- Kirkpatrick C. and Maimbo S.M. (2002), "The implication of the evolving microfinance agenda for regulatory and supervisory policy", *Development Policy Review*, 20(3):293-304.
- Kleinbaum D.G., Kupper L.L., and Muller K.E. (1988), *Applied regression analysis and other multivariable methods* (2nd ed.), PWS Kent Publishing, Boston, MA.
- Koetter M., Kolari J.W. and Spierdijk L. (2012), "Enjoying the quiet life under deregulation: evidence from adjusted Lerner indices for US banks", *Review of Economics and Statistics*, 94, 462-80.
- Labie M. (2004), "Microfinance : un état des lieux", *Mondes en Développement*, 2(126): 9-23.
- Labie, M., Méon, P. G., Mersland, R., and Szafarz, A. (2015). Discrimination by microcredit officers: Theory and evidence on disability in Uganda. *The Quarterly Review of Economics and Finance*, 58, 44-55.
- Latortue, A. (2004), "The Impact of Interest Rate Ceilings on Microfinance", *CGAP Donor Brief*, N°18, 2 pages.
- Maimbo, S. M., and Henriquez Gallegos, C. A. (2014), "Interest rate caps around the world: Still popular, but a blunt instrument", *Policy research working paper*, No. 7070. World Bank Group, Washington.

- Maudo J. and Fernández de Guevara J. (2004), "Factors explaining the interest margin in the banking sectors of the European Union", *Journal of Banking and Finance*, 28, 2259-2281.
- McIntosh C., de Janvry A. and Sadoulet E. (2005), "How rising competition among microfinance institutions affects incumbent lenders", *The Economic Journal*, 115: 987-1004.
- McIntosh and Wydick (2005), "Competition and microfinance", *Journal of Development Economics*, 78 (2), 271-298.
- Ménard C. (1998), "Maladaptation of regulation to hybrid organizational forms", *International Review of Law and Economics*, 18(4): 403-417.
- Mersland R. and Strøm R.O. (2010), "Microfinance mission drift?", *World Development*, 38(1), 28-36.
- Mersland R. and Urgeghe L. (2013), "International debt financing and performance of microfinance institutions", *Strategic Change*, 22(1-2): 17-29.
- Mia M. (2018), "Evolution of market concentration and competition in the microfinance industry in Bangladesh", *International Journal of Social Economics*, 45(6), 989-1007.
- Miller, H. (2013), "Interest rate caps and their impact on financial Inclusion", *Economic and Private Sector Professional Evidence and Applied Knowledge Services (EPS-PEAKS)*, 17 pages.
- MIMOSA – Microfinance Index of Outreach and Saturation (2016), "Cambodia: Multiple borrowing and loan sizes", *Special Circular*, June, 5 pages.
- Mitra, S. K. (2009), "Exploitative microfinance interest rates", *Asian Social Science*, 5(5), 87-93.
- Morduch J. (1999a), "The microfinance promise", *Journal of Economic Literature*, 37(4): 1569-1614.
- Morduch J. (1999b), "The Role of Subsidies in Microfinance: Evidence from the Grameen Bank", *Journal of Development Economics*, 60 (1), 229-248.
- Navajas S., Conning J. and Gonzalez-Vega C. (2003), "Lending technologies, competition and consolidation in the market for microfinance in Bolivia", *Journal of International Development*, 15, 747-770.
- Olivares-Polanco, F. (2005), "Commercializing microfinance and deepening outreach: Empirical evidence from Latin America", *Journal of Microfinance*, 7(1): 47–69.
- Porteous D. (2006), "Competition and microcredit interest rates", *CGAP Focus Note*, N°33, 16 pages.
- Purkayastha D., Tripathy T., and Das B. (2017), "Market structure and state of competition in microfinance: an empirical examination in India", *International Journal of Applied Business and Economic Research*, 15(22), 35-46.
- Reichert P. (2018), "A meta-analysis examining the nature of trade-offs in microfinance", *Oxford Development Studies*, 46(3): 430-452.
- Rhodes S. (1993), "The Herfindahl-Hirschman Index", *Federal Reserve Bulletin*, 79(3), 188-189.
- Rhyne, E., and Christen, R. P. (1999), "Microfinance enters the marketplace", USAID, *Microenterprise Publications*, 37 pages.
- Rigbi, O. (2013), "The effects of usury laws: evidence from the online loan market", *Review of Economics and Statistics*, 95(4): 1238-1248.
- Roberts, P. (2013), "The profit orientation of microfinance institutions and effective interest rates", *World Development*, 41, 120-131.
- Rosengard, J. (2011), "Oversight is a many-splendored thing: choice and proportionality in regulating and supervising microfinance institutions", in *The Handbook of Microfinance*, 159-171, Armendáriz and Labie M. (2011), World Scientific, Singapore.
- Sandberg J. (2012), "Mega-interest on microcredit: are lenders exploiting the poor?", *Journal of Applied Philosophy*, 29 (3): 169-185.
- Schicks J. and Rosenberg R. (2011), "Too much microcredit? A survey of the evidence on over-indebtedness", *CGAP Occasional Papers*, No. 19, 44 pages.
- Schreiner M. (2001), "Seven aspects of loan size", *Journal of Microfinance*, 3(2): 27-47.
- Schreiner M. (2002), "Aspects of outreach: a framework for discussion of the social benefits of microfinance", *Journal of International Development*, 14: 591-603.
- Sinha, S. (2016), "Impact of restrictions on interest rates in microfinance", *Sustainable Development Working Papers Series*, N°45, Manila: Asian Development Bank.

- Van Leuvensteijn M., Bikker J.A., Van Rixel A. and Sørensen C.K. (2007), “A new approach to measuring competition in the loan markets of the euro area”, *European Central Bank Working Paper Series*, No. 768, 37 pages.
- Vanroose A. (2008), “What macro factors make microfinance institutions reach out?”, *Savings and Development*, 32(3), 153-174.
- Vanroose A. and D’Espallier B. (2013), “Do microfinance institutions accomplish their mission? Evidence from the relationship between traditional financial sector development and microfinance institutions’ outreach and performance”, *Applied Economics*, 45: 1965-1982.
- Villegas, D. J. (1989), “The impact of usury ceilings on consumer credit”, *Southern Economic Journal*, 56(1): 126-141.
- Vogelgesang U. (2003), “Microfinance in times of crisis: the effects of competition, rising indebtedness, and economic crisis on repayment behavior”, *World Development*, 31(12): 2085-2114.
- Wagner C. and Winkler A. (2013), “The vulnerability of microfinance to financial turmoil – evidence from the global financial crisis”, *World Development*, 51, 71-90.
- Zetzsche D. and Dewi T.R. (2018), “The paradoxical case against interest rate caps – and: how fintech and regtech resolve the dilemma”, *Law Working Paper Series*, N°2018-003, 37 pages.
- Zinman, J. (2010), “Restricting consumer credit access: household survey evidence on effects around the Oregon rate cap”, *Journal of Banking and Finance*, 34(3): 546-556.

Appendices

Appendix 1. Variables description

Data level	Variable	Description	Source*
Firm-level variables	<i>Average loan size</i>	Gross loan portfolio / number of active borrowers (USD)	MIX Market
	<i>Average loan size adj. GDP p.c.</i>	Average loan size / GDP per capita	MIX Market
	<i>Percentage of female borrowers</i>	Number of loans dedicated to women / total number of loans	MIX Market
	<i>Percentage of rural borrowers</i>	Number of active rural borrowers/total number of active borrowers	MIX Market
	<i>Share of portfolio dedicated to SMEs</i>	Gross loan portfolio dedicated to SMEs/total gross loan portfolio	MIX Market
	<i>Total assets</i>	Total assets, expressed in logarithms	MIX Market
	<i>Breadth of outreach</i>	Number of active borrowers, expressed in logarithms	MIX Market
	<i>Savings</i>	Dummy variable (1 if MFI collects savings, 0 if not)	MIX Market
	<i>Donated equity share</i>	Donated equity / total liabilities	MIX Market
	<i>Sustainable</i>	Dummy variable indicating 1 if the Operational Self-Sufficiency ratio is > 100%, and 0 otherwise	MIX Market
	<i>Operational Expense Ratio (OER)</i>	Operating expenses / gross loan portfolio	MIX Market
	<i>Portfolio at Risk ratio over 30 days (PAR30)</i>	Portfolio at risk for 30 days or more / gross loan portfolio	MIX Market
Country-level variables	<i>Interest rate cap</i>	Dummy variable (1 if cap, 0 if not)	Own search
	<i>Herfindahl-Hirschman Index</i>	Herfindahl-Hirschman Index, based on gross loan portfolio	Own computation
	<i>Number of MFIs in the industry</i>	Number of MFIs in the industry	MIX Market
	<i>GDP growth rate</i>	GDP growth rate	WDI
	<i>Inflation rate</i>	Inflation rate	WDI
	<i>Manufacturing value added</i>	Manufacturing value added in percentage of GDP	WDI
	<i>Percentage of rural population</i>	Percentage of rural population	WDI
	<i>Credit to GDP ratio</i>	Domestic credit / GDP	WDI

* MIX Market = Microfinance Information eXchange; WDI = World Development Indicators (World Bank).

Appendix 2. Use of interest rate caps around the world in 2015-2018

Countries with cap(s) during the whole period	N	Countries without cap during the whole period	N	Countries that imposed caps during the period	N	Countries that removed caps during the period	N
Argentina	13	Bosnia and Herzegovina	16	Cambodia (2017)	66	Zambia (2016)	8
Armenia	24	Burundi	7	Georgia (2017)	17		
Azerbaijan	51	China (de facto)	15	Kenya (2016)	32		
Bangladesh	89	Democratic Republic of Congo	13	Nepal (2017, de facto before)	37		
Benin	27	Costa Rica	25	Sri Lanka (2018, de facto before)	7		
Bolivia	56	Dominican Republic (de facto)	24	Zimbabwe (2016)	2		
Brazil	66	Ethiopia (de facto)	7				
Bulgaria	8	Ghana (de facto)	33				
Burkina Faso	28	Indonesia	19				
Cameroon	25	Jamaica	4				
Chile	9	Jordan	14				
Colombia	73	Kosovo	18				
		Lao PDR (but de facto)	35				
Republic of Congo	1	Liberia	6				
Ecuador	183	Madagascar	26				
Egypt	19	Malawi	3				
El Salvador	10	Mexico	218				
Guatemala	44	Moldova	20				
Honduras	61	Mongolia	15				
India	267	Montenegro	2				
Ivory Coast	12						
Kazakhstan	17	Morocco	17				
Kyrgyz Republic	40	Nicaragua	73				
Mali	14	Nigeria	36				
Myanmar	32	Pakistan (but de facto)	106				
Niger	22	Panama	20				
Paraguay	13	Peru	88				
Poland	1	Philippines	91				
Russia	49	Romania	2				
Senegal	24	Rwanda	13				
		Serbia	2				
Togo	9	Tajikistan	83				
Vietnam	42	Tanzania	11				
		Uganda	13				
		West Bank and Gaza	15				
		Yemen	3				
Total	1329		1093		161		8

Appendix 3. Summary statistics for the variables used for the cost function

Variable	Description	N	Mean	SD
<i>Total cost</i>	Sum of operating and financial expenses (USD)	2183	18 000 000	45 600 000
<i>Gross loan portfolio</i>	Gross loan portfolio (USD)	2591	108 000 000	393 000 000
<i>Cost of labor</i>	Personnel expenses on staff number (USD)	1636	16 747.65	31 436.14
<i>Cost of physical capital</i>	Administrative expenses on net fixed assets	2158	7.278	128.740
<i>Cost of financial capital</i>	Financial expenses on borrowings	2294	0.169	2.885
<i>Output price</i>	Revenues from loans on gross loan portfolio	1979	0.289	0.201
<i>Marginal cost</i>	Marginal cost of lending 1 USD	1535	0.094	0.07
<i>Lerner Index</i>	Price minus marginal cost divided by price	1401	0.662	0.105